Daniil Gerchik

Frankfurt School of Finance & Management, Adickesallee 32–34, 60322 Frankfurt am Main, Germany
d.gerchik@fs.de | daniilgerchik.github.io | linkedin.com/in/daniilgerchik

EDUCATION

Frankfurt School of Finance & Management — **Frankfurt, Germany**
Ph.D. in Finance — September 2020 – Present

- Grade: 1.4 (German scale; 1.0 best)
- Supervisor: Prof. Grigory Vilkov

Yandex School of Data Analysis — **Online**
Machine Learning & Data Science program (M.Sc.-equivalent) — September 2024 – Present

- Coursework: Data Structures & Algorithms, Discrete Math, Python, Machine Learning, Statistics

National Research University "Higher School of Economics" — **Moscow, Russia**
M.Sc. in Financial Economics, in cooperation with LSE, with distinction — September 2015 – June 2017

- GPA: 8.4/10; rank 4/42
- Master's thesis: *Theory and Analysis of the PIN Measure*, under supervision of Prof. Alex Boulatov

National Research University "Higher School of Economics" — **Moscow, Russia**
B.Sc. in Mathematical Economics — September 2011 – June 2015

- GPA: 7.8/10

RESEARCH INTERESTS

Empirical asset pricing; option markets and volatility; machine learning.

WORKING PAPERS

- **When the Chair Speaks: Communication and the FOMC Announcement Premium**

  Abstract: Using option-implied volatility to measure firms' exposure to the FOMC statement and to the Chair's press conference separately, I find that only press-conference exposure is priced in the cross-section of announcement-day returns. Firms most exposed to the press conference earn 108 basis points per event more than the least exposed since Chair Powell's arrival in 2018, while statement exposure is not priced beside it. When the statement stands alone, exposure to it is priced. The premium is mostly earned in the hours before the Chair speaks, and on both hawkish and dovish news. The firms earning it are the fastest-growing. How the Federal Reserve explains a decision, and not only what it decides, shapes which firms bear priced monetary-policy risk.

- **Intraday Volatility-Smile Geometry and Option Returns**

  Abstract: The cross-strike geometry of the implied-volatility smile predicts 30-minute equity option returns. Smiles that are more sharply curved or show larger cross-strike deviations are followed by higher returns; at-the-money options priced rich relative to the fitted smile mean-revert. Yet the same signals carry no information about the underlying stock over the same horizon, and a return decomposition rules out a dominant underlying-move channel. The spreads survive controls for past option returns and standard liquidity proxies. They concentrate in the final half-hour of the session and are largest in stocks where open positions cluster at a small number of strikes, consistent with rebalancing of concentrated inventory toward the close. The cross-strike dimension of the option market has its own short-horizon dynamics, distinct from the behavior of the underlying stock.

- **Factor Dispersions**
  with Vittorio Ruffo, Lorenzo Schönleber, Grigory Vilkov

  Abstract: Dispersion strategies capture the difference in variance dynamics between a basket and its components. Even though smart-beta indices intend to load heavily on a particular factor, factor dispersions based on such baskets are exposed to risks of other factors and idiosyncratic variances. Analyzing factor dispersions through a linear factor model and equicorrelation representations, we recover driving forces behind dispersion dynamics and work out an attribution of a dispersion risk premium. As a balanced combination of systematic and idiosyncratic variance components, dispersion and its risk premium provide signals about future changes in systematic and alpha-based investment opportunities.

WORK IN PROGRESS

- **Dispersion Markets and the Price of Relative News**
  with Vittorio Ruffo, Lorenzo Schönleber, Grigory Vilkov
- **The Digital Euro: News and Narratives**
  with Yevheniia Bondarenko, Vivien Lewis, Christoph Meinerding
- **Bubbles and the Cross-Section of Stock Returns**
  Second-year paper

PRESENTATIONS

- 2026: XXVII Workshop on Quantitative Finance; 3rd Structured Retail Products and Derivatives Conference.
- 2025: 2nd Frankfurt Summer School; PhD Frankfurt School Brownbag.
- 2024: Second Bonn-Frankfurt-Mannheim PhD Conference; PhD Frankfurt School Brownbag.
- 2023: PhD Frankfurt School Brownbag.
- 2022: Frankfurt School Brownbag.

RESEARCH EXPERIENCE

Deutsche Bundesbank — **Frankfurt, Germany**
Research Assistant, Research Centre — January 2026 – Present

Deutsche Bundesbank — **Frankfurt, Germany**
Ph.D. Intern — August 2024 – January 2025

Vienna University of Economics and Business — **Vienna, Austria**
Research Assistant for Prof. Josef Zechner and Maria Chaderina — December 2017 – June 2020

INDUSTRY EXPERIENCE

Deutsche Börse Group – Eurex — **Frankfurt, Germany**
Quantitative Research Intern – Trading Design — April 2025 – September 2025

Sberbank CIB — **Moscow, Russia**
Intern, Structured Products Group — April 2017 – August 2017

Schildershoven Finance B.V. — **Moscow, Russia**
Intern, Fixed Income Research — July 2014 – August 2014

HONORS AND ACHIEVEMENTS

- Ph.D. scholarship and full tuition waiver, Frankfurt School of Finance & Management.
- Full tuition scholarship as winner of the HSE Olympiad for students and graduates in Financial Economics.
- VTB 24 scholarship for academic excellence.
- Bachelor's thesis published in the collection of the best HSE graduate papers, 2015.
- Winner and prizewinner of economics contests and olympiads, including HSE and Interregional School Olympiads.

ADDITIONAL INFORMATION

Technical skills: Python, SQL, R, C++, MATLAB, Git, LaTeX.
Languages: English (fluent), Russian (native), German (A2).
References: Available upon request.